# Chemistry Practical - S1 Integrated MSc

## Experiment 1: Estimation of Sodium Hydroxide using A.R Sodium Carbonate

### What You're Doing (Simple Explanation)

You have two bases (sodium carbonate and sodium hydroxide) and need to figure out how concentrated they are. You'll use hydrochloric acid (a strong acid) to measure their strength by titration. Think of it like finding how much juice is in your drink by mixing it with water until it stops changing color.

### Why This Works (Principle)

When sodium carbonate reacts with hydrochloric acid, they make salt, water, and carbon dioxide. Similarly, sodium hydroxide reacts with hydrochloric acid to make salt and water. By measuring how much HCl is needed, you can figure out the concentration of your unknown solutions.

### Steps to Follow (Procedure)

**Step 1: Make a Standard Solution**

- Weigh out exactly 0.53 g of sodium carbonate ($Na_2CO_3$)
- Dissolve it in water and fill up to the 100 ml mark
- Mix well to spread the concentration evenly

**Step 2: Standardize the HCl (Make Sure It's the Right Strength)**

- Use a pipette to take out 20 ml of your sodium carbonate solution
- Put it in a flask
- Add a few drops of methyl orange indicator (it turns the solution yellow)
- Pour HCl from a burette into the flask slowly, stirring constantly
- Stop when the color changes from golden yellow to pale orange-red
- Write down how much HCl you used
- Repeat this 2-3 times until your results are similar (concordant)

**Step 3: Test the Sodium Hydroxide**

- Take 20 ml of your sodium hydroxide solution
- Add a few drops of phenolphthalein indicator (it turns the solution pink)
- Pour the standardized HCl into it slowly until the pink color disappears completely
- Note how much HCl you used
- Repeat until your results match

### Simple Calculations

From your results, calculate:

- How strong (normal) the HCl is
- How strong (normal) the NaOH is
- Multiply the normality by 40 (the equivalent mass of NaOH) to find the mass of NaOH

## Experiment 2: Estimation of Oxalic Acid

### What You're Doing (Simple Explanation)

You're measuring how much oxalic acid is in a solution using potassium permanganate. This is like using a purple dye that fades when it reacts with acid—when the purple color disappears, you know the reaction is complete.

### Why This Works (Principle)

Potassium permanganate is purple. When it reacts with ferrous salt or oxalic acid, it loses its purple color. You first standardize the permanganate using Mohr's salt, then use it to measure your unknown oxalic acid solution. The permanganate itself is your indicator—when it stops reacting, the solution stays pale pink!

### Steps to Follow (Procedure)

**Step 1: Make a Standard Ferrous Solution (Mohr's Salt)**

- Weigh about 3.92 g of Mohr's salt crystals
- Add these to water with 40 ml of dilute sulfuric acid already added (this prevents damage to the salt)
- Fill up to 100 ml and mix well

**Step 2: Standardize the Permanganate**

- Take 20 ml of your Mohr's salt solution
- Add 20 ml more of dilute sulfuric acid
- Pour potassium permanganate from a burette into it slowly
- Stop when you see the first pale pink color appear (it won't fade away)
- Note how much permanganate you used
- Repeat until results match

**Step 3: Test Your Oxalic Acid**

- Take 20 ml of oxalic acid solution
- Add 20 ml of dilute sulfuric acid

- Gently warm the mixture (make it warm, not hot)
- Pour in the standardized permanganate slowly until it turns pale pink
- Stop when the pink color appears and won't fade
- Note the volume used
- Repeat until results are consistent

### Simple Calculations

From your measurements, calculate:

- The strength of the permanganate
- The strength of the oxalic acid
- Multiply by 63 (equivalent mass of oxalic acid) to find the total mass

## Experiment 3: Total, Permanent and Temporary Hardness of Water

### What You're Doing (Simple Explanation)

Water can be "hard" because it contains calcium and magnesium minerals. Some of this hardness can be removed by boiling (temporary), but some can't (permanent). You'll test water to find both types and the total hardness.

### Why This Works (Principle)

Hard water gets its hardness from calcium and magnesium salts. Temporary hardness comes from bicarbonates—these break down when you boil the water. Permanent hardness comes from chlorides and sulfates—boiling doesn't help. You'll use EDTA (a special chemical that grabs metal ions) to measure both types. EDTA forms complexes with calcium and magnesium, and an indicator dye (Erichrome Black-T) shows when all the metal has been captured.

### Steps to Follow (Procedure)

**Step 1: Measure Total Hardness**

- Take a sample of your water ($V_1$ ml)
- Add ammonia buffer until you can smell ammonia
- Add two drops of Erichrome Black-T indicator (the solution turns wine-red)
- Slowly pour EDTA from a burette into the water while stirring
- Stop when the color changes from wine-red to blue
- Record how much EDTA you used
- Repeat until results match

**Step 2: Measure Permanent Hardness**

- Take a fresh water sample and boil it for about 30 minutes (this removes temporary hardness)
- Cool it down
- Take $V_3$ ml of this boiled water
- Add ammonia buffer and indicator like before
- Titrate with EDTA until the color changes from wine-red to blue
- Record the volume used
- Repeat until consistent

**Step 3: Calculate Temporary Hardness**

- Temporary hardness = Total hardness - Permanent hardness

### Simple Calculations

Using the EDTA concentration (y M):

- For each ml of EDTA, you're removing 0.1y grams of calcium carbonate equivalent
- Calculate for your volumes to find the hardness in ppm (parts per million)
- This tells you if your water is soft, moderately hard, or very hard

## Experiment 4: Transition Temperature and Cryoscopic Constant

### What You're Doing (Simple Explanation)

Some salts (like sodium thiosulfate) have a special temperature where they change from solid to liquid. You'll find this temperature and see how a dissolved chemical changes it.

### Why This Works (Principle)

When sodium thiosulfate gets to its transition temperature, it melts. But if you add a solute (another chemical) to it, the melting point gets lower. By measuring how much lower it gets, you can calculate the cryoscopic constant—a special number that describes how much the solute changes the melting point.

### Steps to Follow (Procedure)

**Step 1: Find the Transition Temperature**

- Put 10-15 g of sodium thiosulfate crystals in a boiling tube
- Insert a thermometer and stirrer through a cork
- Heat the tube in hot water until the solid melts and temperature rises about 5°C more
- Remove the tube and let it cool
- Every 30 seconds, record the temperature
- When it reaches 40°C, add one crystal and stir

- The temperature will suddenly jump up and stay steady for a while
- This steady temperature is your transition temperature!

**Step 2: Repeat with Added Solute**

- Add your test chemical to the salt
- Repeat the heating and cooling process
- Find the new transition temperature
- The difference tells you how much the solute affected it

### Simple Calculations

Use this formula:
$K_f = (M \times W_1 \times \Delta T) / (1000 \times W_2)$

Where:

- M = molar mass of your solute
- $W_1$ = weight of the salt
- $W_2$ = weight of solute added
- $\Delta T$ = difference in transition temperatures

## Experiment 5: Acid-Base Titration by Conductance Measurement

### What You're Doing (Simple Explanation)

Instead of using color changes, you're measuring how well electricity flows through the solution. As acid reacts with base, the conductance (how well it conducts electricity) changes in a predictable way. You'll plot this change to find the exact point where acid and base are perfectly balanced.

### Why This Works (Principle)

HCl has ions that conduct electricity well. When you add NaOH, these ions get used up in the reaction, so conductance goes down. Once all the acid is used up (equivalence point), adding more base makes conductance go back up because the new $OH^-$ ions conduct electricity. By plotting conductance vs. volume of base, you can see exactly where the dip is—that's your equivalence point!

### Steps to Follow (Procedure)

**Step 1: Set Up Your Conductance Meter**

- Prepare 0.1 N KCl by dissolving 0.7456 g KCl in 100 ml water
- Prepare 0.01 N KCl by diluting the above 10 times
- Measure conductance of pure water, 0.1 N KCl, and 0.01 N KCl
- Calculate the cell constant (needed to convert measurements to real values)

**Step 2: Titrate Acid with Base**

- Put 20 ml of 0.1 N HCl in a beaker
- Measure its conductance
- Add 2 ml of NaOH and measure again
- Keep adding 2 ml more each time (4, 6, 8 ml, etc.) and measuring
- Continue until you've added plenty of NaOH

**Step 3: Plot and Find Equivalence Point**

- Make a graph of conductance (y-axis) vs. volume of NaOH (x-axis)
- You'll see the conductance decrease then increase
- The lowest point on the graph is your equivalence point!

### Simple Calculations

At the equivalence point:

- Moles of acid = Moles of base
- Use this to calculate the concentration of your NaOH

## Experiment 6: Viscosity Measurement

### What You're Doing (Simple Explanation)

Viscosity is how "thick" or "gooey" a liquid is. Honey is more viscous than water. You'll measure how fast different liquids flow through a special tube to compare their viscosities.

### Why This Works (Principle)

Liquids with high viscosity flow slowly. When you compare the time it takes two different liquids to flow through the same tube, you can calculate the difference in their viscosities if you know their densities.

### Steps to Follow (Procedure)

**Step 1: Prepare the Viscometer**

- Wash an Ostwald viscometer thoroughly with acetone and dry it
- Pour 10 ml of water into it
- Use the tube attached to it to draw the water up into the upper bulb
- Make sure the liquid level matches the mark above the bulb

**Step 2: Measure Flow Time**

- Release the liquid and let it flow down while starting a stopwatch
- Stop the watch when the liquid reaches the lower mark

- Record the time
- Repeat for other liquids you want to test

**Step 3: Use Your Results**

- Measure the density of each liquid
- Use the formula: viscosity ratio = $(\text{density}_1 \times \text{time}_1)/(\text{density}_2 \times \text{time}_2)$
- Since you know water's viscosity, you can calculate any other liquid's viscosity

## Experiment 7: Determination of Surface Tension

### What You're Doing (Simple Explanation)

Surface tension is what makes water droplets round and lets some insects walk on water. You'll count how many drops of different liquids fall from a tube to measure their surface tension.

### Why This Works (Principle)

A liquid forms a drop when gravity pulls it down and surface tension pulls it up. At a certain point, surface tension can't hold anymore and the drop falls. By counting drops of known volume, you can figure out the surface tension. Comparing one liquid to another (like water) lets you calculate the surface tension of your test liquid.

### Steps to Follow (Procedure)

**Step 1: Prepare the Drop Counter (Stalagmometer)**

- Clean and dry the stalagmometer
- Fill it with water by sucking water up from below
- Adjust the small pinch cork until drops fall at about 10-15 drops per minute
- This is your standard rate

**Step 2: Count Drops of Water**

- Refill the stalagmometer with water
- Let the drops fall from the marked line
- Count how many drops fall until the liquid level drops to the next mark
- Record this number
- Weigh 10 of these drops and calculate the weight of one drop

**Step 3: Test Your Liquid**

- Repeat with your test liquid
- Count and weigh the drops the same way
- Measure the liquid's density

## Simple Calculations

Use this formula:
Surface tension of liquid = (drops of water/drops of liquid) × (density of liquid/density of water) × surface tension of water

The standard surface tension of water at 20°C is about 72.75 mN/m.

## Results Summary

Record all your results for each experiment. Each one tells you something different about the chemicals or properties you're testing. Make sure to report both your measured values and your calculated results!

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1. Lab-manual-S1-PM.docx
2. 1st-experiment.pdf